ENGR 130 Spring 2023

Course Schedule - Updated Jan 22, 2023

Weeks	Date	Ses	Descriptions	Pre class	Release	Due	Assessment
1	1/9	1	Course overview and Design Challenge 1		HW1		
	1/11	2	Images of Engineering - Kicking off EYC activities EXCEL 1 - Analyzing system performance with data (introduction to Data Acquisition System)	PCV			
	1/13	3	EXCEL 2 - Importing data - Descriptive Stat and Histograms Design Lab 1: Reaction time study	PCV+Quiz	HW 2	HW1	
2	1/16	1	**Martin Luther King, Jr Holiday - No class**				
	1/18	2	EXCEL 3 - Statistics - Introduce Project 1 (introducing telemetry system) Design Lab 2: characterize acceleratometer	PCV; Install Mu for 2.1			
	1/20	3	Teams and Teaming CoC; Models of Design and process (Systems Thinking)		Project 1	HW 2	
3	1/23	1	Making Sense of System: Systems thinking and functional block diagrams (vs component block diagrams)		HW 3		
	1/25	2	Flow charting and computational models		EYC-1		
	1/27	3	Python 1 - Getting Started with Spyder; Introduction to programming and Python	PCV+Quiz; Install Python	HW 4	HW 3	CFU
4	1/30	1	Project 1 Demonstrations;		Project 3		
	2/1	2	Python 2 - Data Collection Types/Conditionals and User Defined Functions	PCV+Quiz	EYC-2		
	2/3	3	Design Lab 4 - (Leveling system) Characterizing Sensors: basics of sensors (internal: Use accel for tilt sensor and Auto tone,		HW 5	HW 4+CoC	
5	2/6	1	Project 1 Reflection Python 3 - Loops and Plotting	PCV+Quiz		EYC-1	CFU
	2/8	2	Exam 1				Exam 1
	2/10	3	Python 3 - Loops		HW 6	HW 5; Project 3 Proposal	
6	2/13	1	Project Management				
	2/15	2	Ideation, Brainstorming and Innovative thinking (brainstorming and Morphcharts)	PCV+Quiz			CFU
	2/17	3	Python 4 - File I/O		HW 7	HW 6	
7	2/20	1	Design Lab 5 - External Sensors, servos and motors				
	2/22	2	Decision making; PPM;Estimation/Approx.				
	2/24	3	Project 2 development - Milestone 1 - remote control		HW8; Project 2	HW 7	
	2/27	1	Project 2 - Milestone 2: Algorithm options (Day 2)				

8	3/1	2	Project 2 - Milestone 3: Work Day (Day 3)				
	3/3	3	Project 2 Demonstrations			HW 8	
9	3/6	1	Exam 2				Exam 2
	3/8	2	MATLAB 1 Introduction	PCV+Quiz	HW 10	Project 2 Report Due	
	3/10	3	Linear systems/ Method of selected points. Goodness of fit.		Project 4 Release	HW 9	
	3/13	1	**Spring Break Week - No Classes**				
	3/15	2					
	3/17	3					
10	3/20	1	Function Discovery 1 (non linear systems)				
	3/22	2	MATLAB 2 - Arrays, Data Load, and Plotting (histograms)	PCV+Quiz	HW 11		CFU
	3/24	3	Function Discovery 2 Method of Least Squares.			HW 10	
11	3/27	1	Project 3 work day - Design review with Peer Advisory Board				
	3/29	2	MATLAB 3 - Conditionals and loops	PCV+Quiz	HW 12	Project 4 Proposal	CFU
	3/31	3	MATLAB Modeling - Project 3 Work day			HW 11	
12	4/3	1	Project 3 work day				
	4/5	2	MATLAB 4 - File IO	PCV+Quiz	HW 13		CFU
	4/7	3	Project 3 Demonstration - Tech Fair			HW 12	
13	4/10	1	MATLAB 5 - Regression	PCV+Quiz		Project 3 Video Presentation	
	4/12	2	Project 3 Presentation - (Presentation done in class with Peer Review)			Project 3 Report	CFU
	4/14	3	Project 4 Workday			HW 13	
14	4/17	1	Project 4 Workday		EYC-3	EYC-2	
	4/19	2	Exam 3				Exam 3
	4/21	3	Project 4 Workday				
15	4/24	1	Project 4 Presentation				
	4/26	2	Project 4 Presentation				
	4/28	3	Adjournment - Project 4 report due - Review final videos of Best of Project 3			Project 4 Report; EYC-3	
16	5/1 - 5/6		**FINALS WEEK**				